

Transport Layer:-

- The **Transport Layer** is a crucial component of the OSI and TCP/IP models, responsible for **process-to-process communication** by ensuring data is transferred between applications running on different hosts.
- It provides **end-to-end communication, reliability, and flow control** depending on the protocol used.

Responsibilities of a Transport Layer

- The Process to Process Delivery
- End-to-End Connection between Hosts
- Multiplexing and Demultiplexing
- Congestion Control
- Data integrity and Error correction
- Flow control

◆ Transport Layer Protocols:

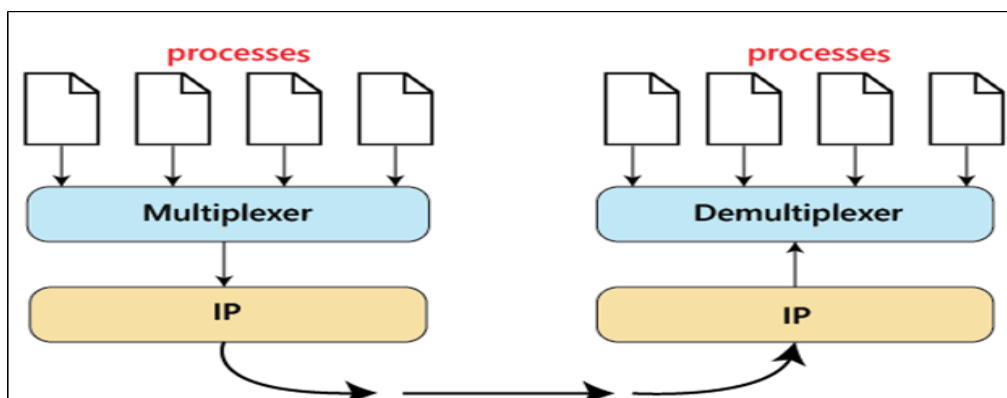
- **TCP** (Reliable, ordered, connection-oriented).
- **UDP** (Fast, connectionless, no delivery guarantee).
- **SCTP** (Supports multiple streams, used in telecom).

◆ Multiplexing & Demultiplexing:

- **Multiplexing:** Multiple apps can send data over the network at the same time.
- **Demultiplexing:** The transport layer delivers data to the correct application using port numbers.

Process-to-Process Communication in the Transport Layer

- Process-to-process communication ensures that data sent from an application on one device reaches the correct application on another device.
- This is essential because multiple applications (processes) can be running simultaneously on both sender and receiver devices.



Transport Layer Protocols & Congestion Control

1. User Datagram Protocol (UDP)

- **Connectionless, fast, and unreliable.**
- No error checking, retransmission, or order guarantee.
- Used in: **Live streaming, VoIP, DNS, gaming.**

2. Transmission Control Protocol (TCP)

- **Connection-oriented, reliable, and ordered.**
- Uses **three-way handshake (SYN, SYN-ACK, ACK)** for connection.
- Ensures data integrity via **error checking, retransmission, and acknowledgments.**
- Used in: **Web browsing (HTTP/HTTPS), Email, File transfer (FTP).**

3. Stream Control Transmission Protocol (SCTP)

- Combines **features of both TCP and UDP.**
- Supports **multi-homing** (can switch networks without disconnecting).
- Uses **multiple streams** for better reliability.
- Used in: **Telecom signaling (SS7 over IP), financial transactions.**

Congestion Control

- Congestion occurs when network traffic exceeds capacity, leading to packet loss and delays.
- TCP includes congestion control mechanisms to avoid overwhelming the network:
 - **Slow Start:** Gradually increases the amount of data sent.
 - **Congestion Avoidance:** Reduces transmission rate when packet loss is detected.
 - **Fast Retransmit & Recovery:** Detects lost packets and resends them quickly without reducing speed too much.

♦ TCP Congestion Control Mechanisms:

1. **Slow Start** – Increases transmission rate gradually.
2. **Congestion Avoidance** – Adjusts speed based on network conditions.
3. **Fast Retransmit & Fast Recovery** – Quickly resends lost packets without slowing down too much.

♦ UDP & SCTP:

- **UDP** has **no congestion control**, making it fast but risky.
- **SCTP** includes congestion control similar to TCP.

Effects of Congestion Control :-

- ✓ **Keeps the network stable** – Prevents overload and crashes.
- ✓ **Reduces delays & data loss** – Makes data transfer faster and smoother.
- ✓ **Improves speed** – Allows more data to be sent efficiently.
- ✓ **Shares network fairly** – No one user takes up all the bandwidth.
- ✓ **Better experience** – Websites, videos, and apps run smoothly.
- ✓ **Prevents network breakdown** – Stops traffic spikes from making the network unusable.

Congestion Control Algorithms :-

Congestion control helps prevent network overload by managing data flow. Two main algorithms used are:

1. Leaky Bucket Algorithm

- Works like a bucket with a small hole—water (data) leaks at a fixed rate, no matter how fast it's poured in.
- Ensures a steady data flow, preventing sudden bursts.
- If the bucket (buffer) is full, extra data is lost.
- Used for **traffic shaping** but can be inefficient in using bandwidth.

2. Token Bucket Algorithm

- A bucket stores **tokens**, each allowing a data packet to be sent.
- If tokens are available, data can flow freely. No tokens = no data transmission.
- Allows **bursty traffic** when enough tokens are stored.
- More flexible than **Leaky Bucket**, as it doesn't drop excess data.

Quality of Service (QoS) :-

- **Quality of Service (QoS)** refers to techniques used to ensure efficient and reliable network performance by prioritizing important traffic and managing bandwidth.
- **Quality-of-service (QoS)** refers to traffic control mechanisms that seek to differentiate performance based on application or network-operator requirements or provide predictable or guaranteed performance to applications, sessions, or traffic aggregates.

Why is QoS Important?

- Ensures smooth video calls, gaming, and streaming.
- Prevents network congestion and delays.
- Helps critical applications (e.g., emergency calls) get priority.

Types of Quality of Service (QoS)

Stateless Solutions

- Routers **do not track individual traffic flows**.
- **Scalable & robust** but **no performance guarantees** (e.g., no delay control).
- Example: **Best Effort Service** (no priority for any traffic).

Stateful Solutions

- Routers **track traffic flows** to ensure **better QoS**.
- Provides **guaranteed services, protection, and efficient resource use**.
- **Less scalable** due to high resource usage.
- Example: **Integrated Services (IntServ)** (reserves bandwidth for critical traffic).

Key QoS Factors / QoS Parameters:-

- ✓ **Bandwidth** – Amount of data the network can handle.
- ✓ **Latency** – Delay in data transmission (lower is better).
- ✓ **Jitter** – Variation in packet arrival times (affects calls/streams).
- ✓ **Packet Loss** – Data packets lost due to congestion (causes disruptions).

Benefits of QoS

- Improved Performance for Critical Applications
- Enhanced User Experience
- Efficient Bandwidth Utilization
- Increased Network Reliability
- Compliance with [Service Level Agreements \(SLAs\)](#)
- Reduced Network Costs
- Improved Security
- Better Scalability

Models to Implement QoS :-

1.Integrated Services (IntServ)

- **Provides QoS guarantees for individual applications.**
- **Uses resource reservation (routers track and allocate resources).**
- **Admits or denies traffic based on available resources.**

2.IntServ QoS Components

- **Resource Reservation** (setup, traffic control, admission control).
- **QoS-Sensitive Scheduling** (e.g., Weighted Fair Queuing - WFQ).
- **QoS-Sensitive Routing** (e.g., Quality Shortest Path First - QSPF).
- **Packet Discard Strategy** (prioritizes important packets).

3. RSVP (Resource Reservation Protocol)

- Used for signaling QoS reservations across the network.
- Receiver-initiated (reserves bandwidth via RESV messages).
- Soft state (needs to be refreshed periodically).

4. Call Admission Control

- Session declares QoS needs (delay, packet loss, etc.).
- **R-Spec**: Defines required QoS (e.g., delay, loss limits).
- **T-Spec**: Describes traffic pattern (e.g., burstiness).
- Routers accept or reject traffic based on available resources.

5. Differentiated Services (DiffServ)

- More scalable than IntServ (does not track individual flows).
- Uses priority-based traffic classes instead of strict reservations.
- Simpler signaling (focuses on broader traffic categories).

QoS improving techniques: Leaky Bucket and Token Bucket algorithm.

1. Leaky Bucket Algorithm.

The Leaky Bucket Algorithm is a network traffic shaping technique that controls data flow to prevent congestion by ensuring a constant output rate, regardless of bursty input traffic.

How It Works:

- Imagine a bucket with a hole at the bottom.
- Water (data packets) is added at varying rates but leaks at a constant rate.
- If the bucket overflows, excess water (packets) is discarded.

Algorithm Steps (Variable-Length Packets):

1. Initialize a counter to n (max burst size) at each clock tick.
2. Check the packet at the queue's head:
 - If its size is greater than n, wait.
 - Else, send the packet and decrease the counter by its size.
3. Repeat until the counter is exhausted.
4. Reset the counter and restart the process.

Key Points:

- Smooths bursty traffic by ensuring a steady data flow.
- Prevents network congestion by limiting excessive traffic.
- Uses a FIFO queue to process packets efficiently.

Token Bucket Algorithm:-

The Token Bucket Algorithm is a network traffic shaping technique that allows bursts of data while ensuring overall traffic conforms to a defined rate.

How It Works:

- Imagine a bucket that fills with tokens at a fixed rate.
- Each packet requires a token to be sent.
- If enough tokens are available, the packet is transmitted.
- If no tokens are left, the packet must wait until new tokens arrive (or may be dropped if the buffer is full).

Algorithm Steps:

1. Tokens are added to the bucket at a constant rate.
2. If a packet arrives:
 - Check if there are enough tokens for the packet's size.
 - If yes, send the packet and deduct the tokens.
 - If no, the packet waits (or is discarded if the queue is full).
3. The bucket has a maximum capacity; excess tokens are discarded if the bucket is full.

Key Features:

- Allows bursts of traffic if enough tokens are available.
- Ensures controlled transmission over time.
- More flexible than the Leaky Bucket, as it permits bursts while still enforcing an average rate.

Token Bucket Algorithm	Leaky Bucket Algorithm
It depends on tokens.	It does not depend on tokens.
If bucket is full, token is discarded but not the packet.	If bucket is full, then packets are discarded.
Packets can only transmit when there are enough tokens.	Packets are transmitted continuously.
Allows large bursts to be sent at faster rate. Bucket has maximum capacity.	Sends the packet at a constant rate.

Application Layer Protocols & Concepts

The **Application Layer** is the **topmost layer** in the **OSI model** and provides network services directly to user applications. It facilitates communication between users and the network through various protocols and security mechanisms.

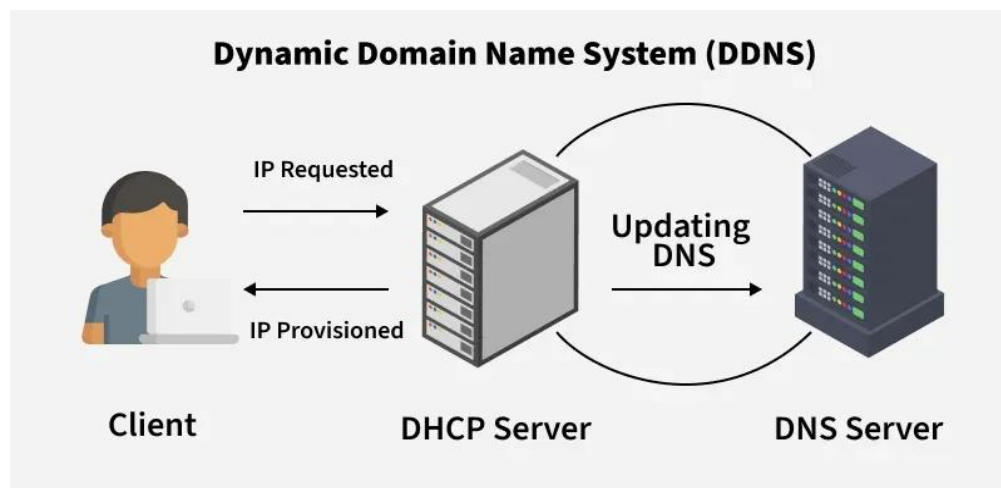
Domain Name Space (DNS) :

The Domain Name System (DNS) translates human-readable domain names (e.g., www.google.com) into machine-readable IP addresses (e.g., 142.250.190.14), enabling internet communication

- It enables computers to locate and communicate with each other on the internet.
- Functions as a hierarchical, distributed database.
- Queries pass through multiple levels:
 - Root server
 - Top-Level Domain (TLD) server
 - Authoritative server (stores the specific IP address).
- Ensures seamless website access using easy-to-remember names instead of numerical IP addresses.

Dynamic DNS (DDNS):

- Dynamic Domain Name System (DDNS) is a method of automatically updating a name server in the Domain Name Server (DNS) often in real-time, with the active DDNS configuration of its configured hostnames, addresses or other information.
- Dynamic DNS (DDNS) is a service that automatically updates the DNS records when a device's IP address changes, ensuring continuous domain-to-IP mapping.



Common DDNS Providers:

- **No-IP** (noip.com)
- **DynDNS** (dyndns.com)
- **DuckDNS** (duckdns.org)

Advantages of Dynamic DNS (DDNS):

- ✓ Automatically updates the domain with the latest IP address.
- ✓ Saves money by avoiding the need for a static IP address.
- ✓ Allows remote access to home servers, CCTV cameras, and other devices.
- ✓ Easy to set up on most routers and devices.
- ✓ Useful for IoT and smart home devices that require remote control.
- ✓ Ensures a domain always points to the correct IP, preventing downtime.

Disadvantages of Dynamic DNS (DDNS):

- ✗ Can be vulnerable to cyberattacks if not properly secured.
- ✗ Relies on third-party DDNS providers, which may shut down or change policies.
- ✗ Not ideal for large businesses that require stable, static IPs.
- ✗ Some ISPs may block DDNS services or restrict home server hosting.
- ✗ Updates may have slight delays when the IP address changes.

TELNET

- TELNET (TELEcommunication NETwork) is a **command-line protocol** used for remote login to another computer over a network.
- TELNET is a type of protocol that enables one computer to connect to the local computer.
- It is used as a standard TCP/IP protocol for virtual terminal service which is provided by ISO.
- The computer which starts the connection is known as the local computer.

Advantages of TELNET:

- ✓ Allows remote access to computers and servers over a network.
- ✓ Enables administration and troubleshooting of network devices.
- ✓ Supports multi-user environments where multiple users can access the same system.
- ✓ Simple and lightweight protocol with minimal resource usage.
- ✓ Can be used for testing and debugging network services.
- ✓ Works across different operating systems without requiring a GUI.

Disadvantages of TELNET:

- ✗ Transmits data, including passwords, in plaintext, making it highly insecure.
- ✗ Vulnerable to eavesdropping, man-in-the-middle attacks, and unauthorized access.
- ✗ Lacks encryption, making it unsuitable for sensitive data transmission.
- ✗ Blocked by many modern networks and replaced by SSH for security reasons.
- ✗ Limited functionality for file transfers and advanced system management.

Email (Electronic Mail)

Email is a method of exchanging digital messages over a network, allowing users to send and receive text, images, documents, and other attachments.

It is widely used for communication in both personal and professional environments.

Advantages of Email:

- ✓ Fast and efficient communication across the world.
- ✓ Allows sending of text, images, documents, and multimedia files.
- ✓ Supports multiple recipients through CC (carbon copy) and BCC (blind carbon copy).
- ✓ Accessible from any device with an internet connection.

Disadvantages of Email:

- ✗ Vulnerable to spam, phishing attacks, and malware.
- ✗ Requires an internet connection to send or receive messages.
- ✗ Can be misinterpreted due to lack of tone and facial expressions.
- ✗ Overloaded inboxes can make it difficult to manage important emails.
- ✗ File attachment size limitations may restrict large file sharing.

Email Protocols :

Email protocols define how emails are sent, received, and stored between mail servers and clients. The three main email protocols are **SMTP, POP3, and IMAP**, each serving a different function in email communication.

1. SMTP (Simple Mail Transfer Protocol):-

- ✓ Used for sending emails from a client to a mail server.
- ✓ Transfers emails between mail servers.
- ✓ Works efficiently for outgoing mail delivery.
- ✓ Supported by all email clients and servers.
- ✓ Uses port **25, 587 (with encryption), or 465 (for SSL)**.
- ✗ Cannot retrieve or store emails; only sends them.
- ✗ Lacks built-in security unless combined with encryption (SSL/TLS).
- ✗ Requires additional protocols like POP3 or IMAP for receiving emails.

2. POP3 (Post Office Protocol v3)

- ✓ Downloads emails from the mail server to the local device.
- ✓ Frees up server space by storing emails locally.
- ✓ Allows offline access to emails once downloaded.
- ✓ Uses port **110 (default) or 995 (SSL/TLS encryption)**.

- ✗ Emails are deleted from the server after download (unless configured otherwise).
- ✗ Cannot access emails from multiple devices easily.
- ✗ No synchronization between email clients and the server.

3. IMAP (Internet Message Access Protocol)

- ✓ Keeps emails stored on the server, allowing access from multiple devices.
- ✓ Synchronizes email status (read, unread, deleted) across devices.
- ✓ Supports folder management for organizing emails.
- ✓ Uses port **143 (default) or 993 (SSL/TLS encryption)**.
- ✗ Requires an internet connection to access emails.
- ✗ Uses more server storage compared to POP3.
- ✗ Can be slower due to real-time synchronization with the server.

File Transfer Protocol (FTP)

- FTP is a standard network protocol used for transferring files between a client and a server over the internet or a local network.
- It allows users to **upload, download, and manage files** on a remote server.

Advantages of FTP:

- ✓ Efficient for transferring large files between computers.
- ✓ Supports **resume functionality**, allowing interrupted transfers to continue.
- ✓ Enables batch file transfers, making it useful for bulk uploads/downloads.
- ✓ Provides authentication via username and password for secure access.
- ✓ Can be automated using scripts for scheduled file transfers.
- ✓ Available on multiple operating systems and supported by various FTP clients.

Disadvantages of FTP:

- ✗ Transmits data in plaintext, making it vulnerable to hacking and eavesdropping.
- ✗ Requires **FTP client software** for full functionality, unlike HTTP-based transfers.
- ✗ Firewall restrictions may block FTP connections due to multiple port usage.
- ✗ Lacks strong security unless encrypted versions like **FTPS (FTP Secure) or SFTP (SSH File Transfer Protocol)** are used.
- ✗ Can be complex to configure for beginners, especially with permissions and encryption settings.
- ✗ Slower compared to modern cloud-based file-sharing services.

Similarities Between FTP and SMTP

- ✓ Both are application layer protocols in the OSI model.
- ✓ Both use TCP for reliable data transmission.
- ✓ Both require user authentication for secure access.

- ✓ Both can be secured using encryption (FTPS/SFTP for FTP, TLS/SSL for SMTP).
- ✓ Both are widely used for data exchange over the internet.

Difference Between FTP and SMTP

FTP	SMTP
Used for transferring files between a client and a server.	Used for sending emails between mail servers.
Supports uploading, downloading, and managing files.	Only transmits emails and does not store them permanently.
Handles all types of files (text, images, videos, etc.).	Designed specifically for email messages and attachments.
Works with both client-server and peer-to-peer communication.	Works strictly on a client-server model.
Uses port 21 for control and port 20 for data transfer.	Uses port 25, 587, or 465 for email transmission.
Requires login credentials for file access.	Authentication is mainly needed for sending emails.
Maintains an active session during file transfer (stateful).	Each transaction is independent (stateless).
Not secure unless FTPS or SFTP is used.	Needs TLS/SSL encryption for secure email transmission.

WWW (World Wide Web)

- A collection of interconnected web pages and resources
- Accessed through web browsers like Chrome, Firefox, and Edge
- Uses URLs to locate resources on the internet
- Based on protocols like HTTP and HTTPS for communication
- Supports multimedia content, including text, images, and videos
- Enables online services like e-commerce, social media, and news

HTTP (HyperText Transfer Protocol)

- A protocol used for communication between web browsers and servers
- Uses request-response model (GET, POST, PUT, DELETE)
- Operates on port 80 (HTTP) and port 443 (HTTPS)

- Lacks built-in security, leading to the development of HTTPS
- Stateless protocol, meaning each request is independent
- Used for loading web pages, APIs, and other internet resources

SNMP (Simple Network Management Protocol)

- Used for monitoring and managing network devices like routers and switches
- Works on UDP ports 161 (queries) and 162 (traps)
- Uses a centralized management model with SNMP agents and managers
- Collects network performance data and detects faults
- Operates in different versions (SNMPv1, SNMPv2, SNMPv3)
- Commonly used in enterprise networking for monitoring traffic and device health

Difference Between HTTP and FTP

HTTP (HyperText Transfer Protocol)	FTP (File Transfer Protocol)
Transfers web pages and resources	Transfers files between computers
Uses port 80 (HTTP) and 443 (HTTPS)	Uses ports 20 (data) and 21 (control)
Stateless (each request is independent)	Stateful (maintains a session)
Transfers web pages, images, and multimedia	Transfers files in bulk or individually
Follows a request-response model	Uses control and data connections for file exchange
Supports HTTPS for encryption	Requires FTPS or SFTP for secure transfers
Used for web browsing, APIs, and web applications	Used for uploading/downloading files, backups, and data sharing

Similarity between HTTP and FTP

- Both use TCP as the underlying transport protocol for reliable data transmission.
- Both involve client-server architecture for communication.
- Both enable data transfer between remote systems.
- Both can be secured using encryption protocols (HTTPS for HTTP, FTPS/SFTP for FTP).

Bluetooth :

- Wireless communication technology for short-range data exchange
- Uses radio waves in the 2.4 GHz frequency range
- Supports low-power consumption, making it ideal for mobile devices
- Commonly used for connecting peripherals like headphones, keyboards, and smartwatches

- Operates within a limited range (typically 10-100 meters)
- Supports different versions (e.g., Bluetooth 4.0, 5.0, 5.3) with improved speed and range
- Enables secure pairing through encryption and authentication mechanisms
- Used in IoT, healthcare devices, file sharing, and hands-free communication

Firewalls :

- A security system that monitors and controls incoming and outgoing network traffic
- Acts as a barrier between a trusted internal network and untrusted external networks
- Can be hardware-based, software-based, or cloud-based
- Uses predefined rules to allow or block specific traffic
- Protects against cyber threats like unauthorized access, malware, and hacking attempts
- Supports different types such as packet filtering, stateful inspection, and proxy firewalls
- Often used in combination with other security measures like antivirus and intrusion detection systems
- Essential for securing enterprise networks, personal devices, and cloud environments

Cryptography :

- The practice of securing communication and data through encryption techniques.
- Ensures **confidentiality, integrity, authentication, and non-repudiation** of information.
- Used in applications like secure messaging, digital signatures, and online banking.

Types of Cryptography

- **Symmetric Cryptography** – Uses a single key for both encryption and decryption (e.g., AES, DES).
- **Asymmetric Cryptography** – Uses a public key for encryption and a private key for decryption (e.g., RSA, ECC).
- **Hashing** – Converts data into a fixed-length string to verify integrity (e.g., SHA-256, MD5).

Common Cryptographic Techniques

- **Encryption & Decryption** – Converting plaintext into ciphertext and vice versa.
- **Digital Signatures** – Verifies the authenticity of messages or documents.
- **Steganography** – Hiding data inside other media, like images or audio files.

- **Key Management** – Securely generating, storing, and distributing cryptographic keys.

Applications of Cryptography

- Secure online transactions (e.g., SSL/TLS in HTTPS).
- Password protection and authentication systems.
- Data protection in cloud storage and communication.
- Digital rights management (DRM) for media content security.